

# Labor Economics Empirical Assignment

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# 1 Replication Autor Katz and Kearney (2008)

## “Trends in U.S. Wage Inequality: Revising the Revisionists”

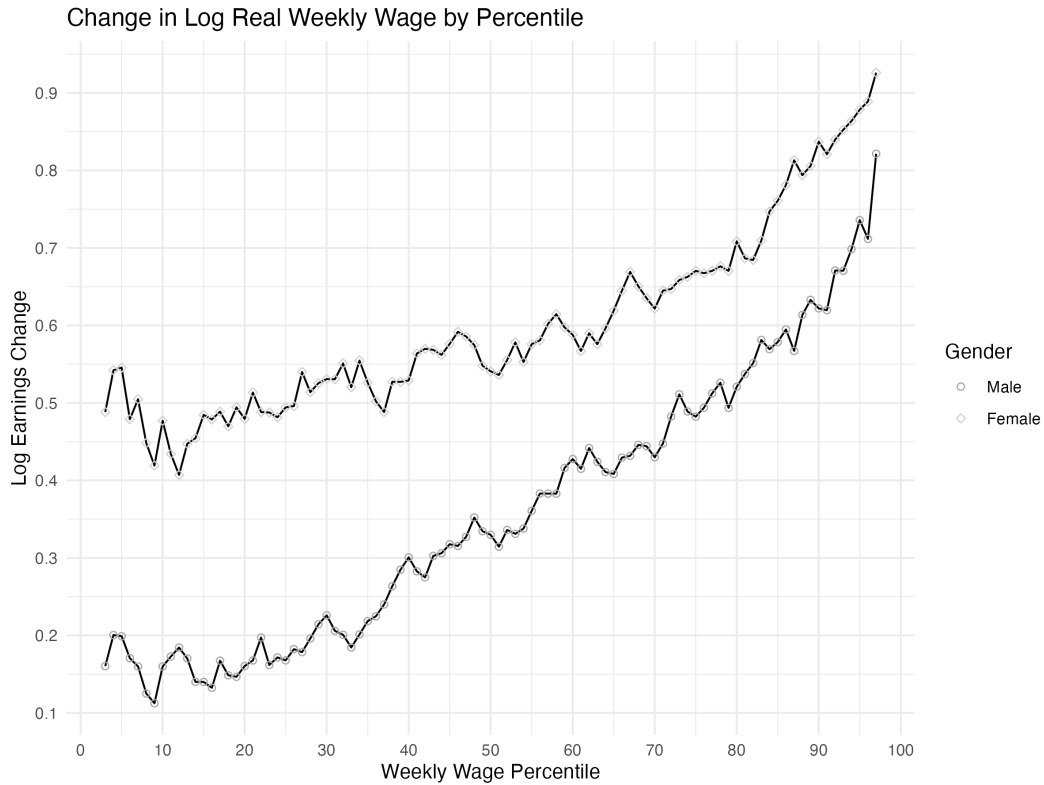


Figure 1: Change in Log Real Weekly Wage by Percentile, Full Time Workers, 1963–2005

Figure 1 shows the overall trend in inequality by looking at the difference in earnings between the year 1963 and the year 2005. The earnings distribution is broken down into percentiles, therefore by analysing how much each percentile has grown between these two years, a growth trend between the different percentiles becomes immediately apparent. Plotted are the changes in log earnings for each percentile, for both men and women. Both men's and women's log earnings grew at each percentile between 1963 and 2005, with women showing larger differences throughout than men. Overall, growth differences are also higher for higher percentiles, showing an almost linear upward trend in earnings change per percentile. This supports Autor, Katz

and Kearney's hypothesis that wage inequality has grown steadily over the last four decades, rather than being an episodic event as the revisionists claim.

In fact, the 90th percentile has gained 47 log points more than the 10th percentile for men. For women this analysis shows 37 log points. It is also noteworthy that, although the change in log earnings for men is at a lower level, the differences are more pronounced in relative terms for higher percentiles from the 30th percentile upwards.

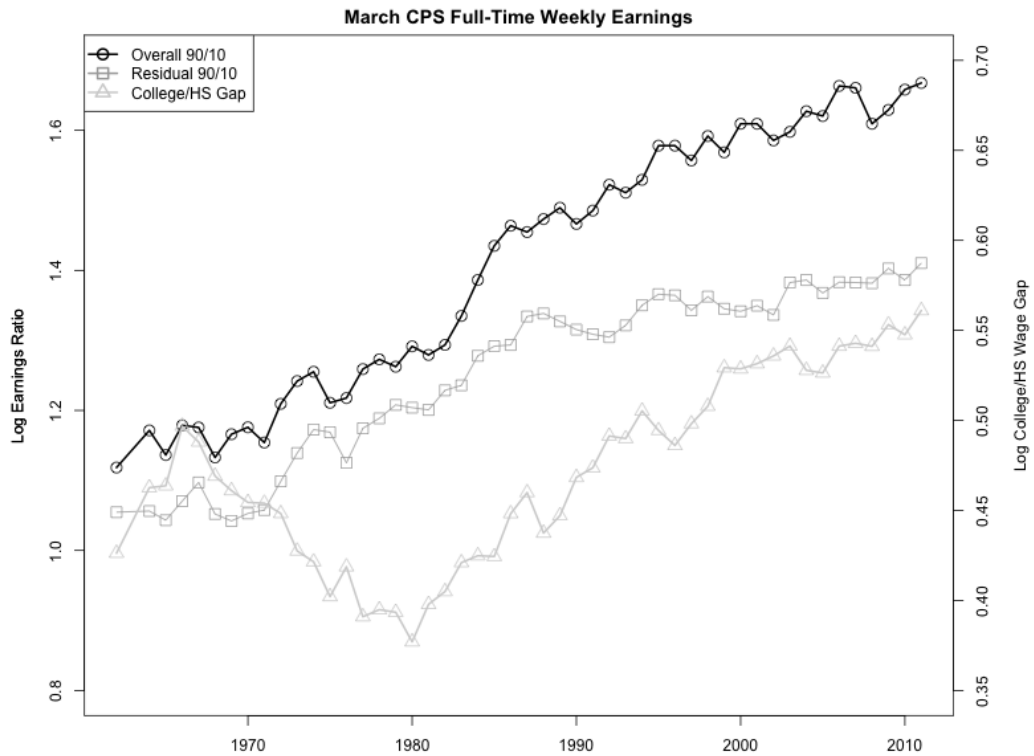


Figure 2: Three Measures of Wage Inequality: College/HighSchool Premium, Male 90/10 Overall Inequality, and Male 90/10 Residual Inequality

Figure 2 looks at the evolution of inequality in three different measures for the period 1963-2005: overall inequality represented by the 90/10 log wage percentile difference, the 90/10 residual difference and the college/high school wage premium. Residual wage inequality is the residual part of remaining after regressing on education, and therefore represents the unexplained part for the wage inequality. It is again plotted as the differential of the 90th and 10th percentile. The college/high school wage premium regresses the

college/high school wage log wages for the subgroups experience, education, gender (interaction of experience and education) to compare observations that differ only in education. Both the residual and the overall measure of inequality rise fairly consistently over the observed period, with the overall measure unsurprisingly at a higher level than the residual. In our analysis, the residual rises at a significantly higher level (but at a similar rate) than in the paper. We explain this discrepancy by our smaller dataset, which omits several potentially explanatory variables, thus increasing the unexplained part of the model, which here is our residual. The lowest, but also rising inequality measure (at least since 1981) is the college/high school ratio. Although it rises until 1973, it falls quite sharply towards the 1980s and plateaus there for four years before assuming the same trend as the other two measures.

## 2 Replication Autor and Dorn (2013), “The Growth of Low-Skill Service Jobs and the Polarization of the US Labor Market

In Figure 3, the smoothed changes in employment and hourly wages between 1980 and 2005 are presented. The occupations are divided into skill percentiles, ranked by the 1980 occupational mean wage, for comparison.

In Panel A, the results highlight a positive change in the upper tail, supporting the conventional theory of job polarization due to technological advancements, which favor high-skill, non-routine jobs. Notably, there is also an increase in employment in the lower tail, indicating that technological progress has not only boosted the demand for non-routine jobs but has also led to a rise in manual, low-skill jobs. Compared to the original paper, the observed changes are smaller overall, with a notable increase around the 50th skill percentile. This discrepancy might stem from differences in the dataset and the calculation methods used, which could affect the magnitude and distribution of observed changes.

In Panel B, the changes in real log hourly wages across skill percentiles are displayed. Over the years, wages have increased across all skill percentiles and job types. However, the strongest increases are observed in the lower and upper tails of the distribution. This pattern suggests that non-routine jobs, both high-skill and manual, have become more valuable, while routine jobs have seen comparatively smaller wage growth due to technological advancements that make them easier to perform. Compared to the original paper, the drop in wages at the lower tail is not observed here, meaning that low skilled jobs had an similar strong increase in wage as high skilled jobs.

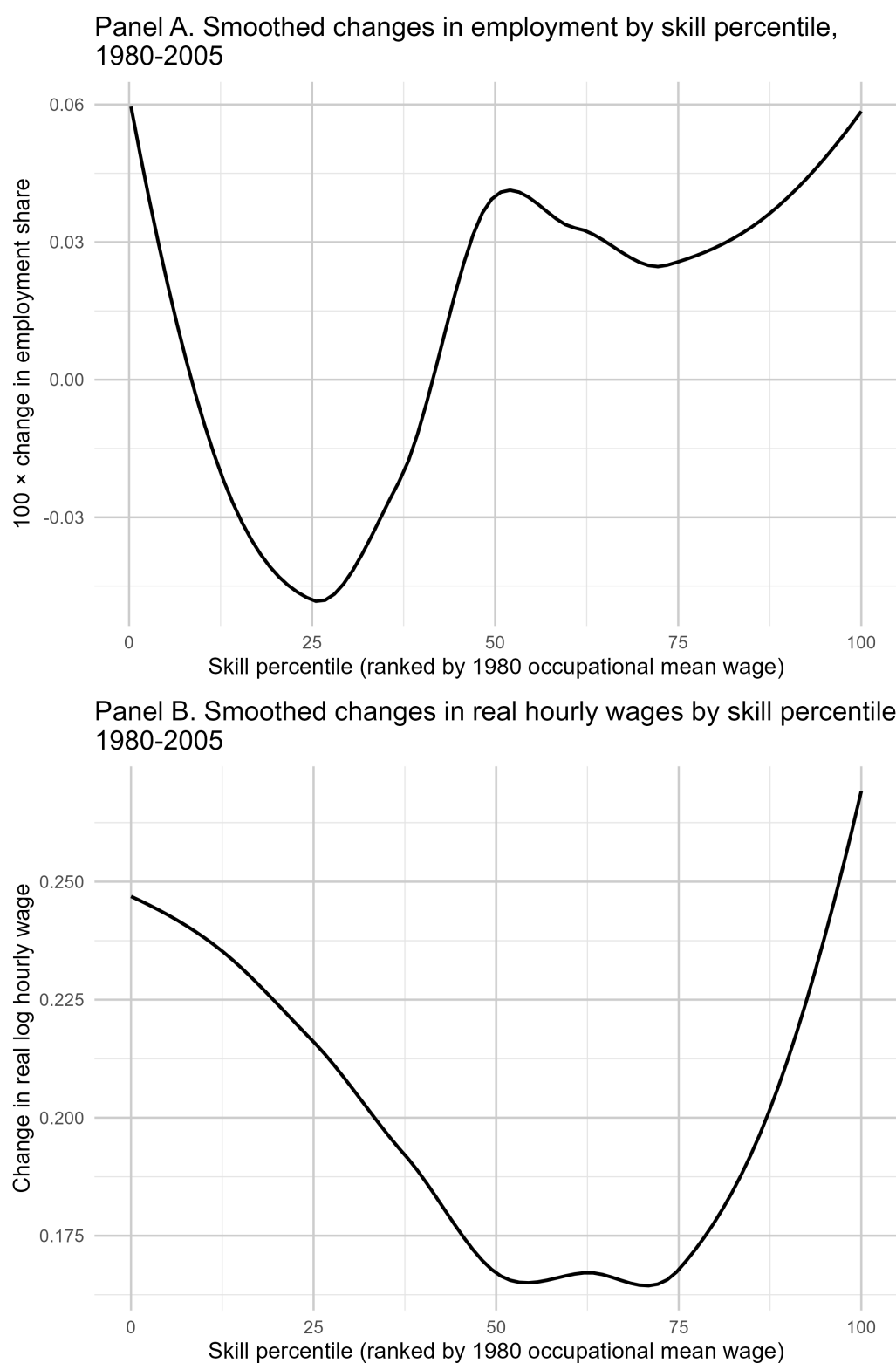


Figure 3: Smoothed Changes in Employment and Hourly Wages, 1980–2005

Figure 4 revisits the analysis in Figure 3 but introduces adjustments: Panel A includes the change in employment share when service occupations are held constant at their 1980 levels, while Panel B shows the change in real log hourly wages when wage growth for service occupations is set to zero.

In Panel A, an unexpected divergence from the results in Autor and Dorn is observed. For skill percentiles between 50 and 100, the two curves (observed and adjusted) are nearly identical. This similarity can be attributed to the fact that service jobs, which impact the curve significantly, are predominantly concentrated in the lower skill percentiles. However, for the lower skill percentiles, the adjusted curve deviates in the opposite direction compared to the original paper. Specifically, the data suggest that the employment share for low-skilled jobs in 2005 is higher than in 1980. This discrepancy can likely be explained by differences in the dataset. In this dataset, there are more total workers in 1980 than in 2005. When service employment from 1980 is carried over to 2005, the total number of workers increases significantly, which disproportionately inflates the share of lower-skilled jobs. This results in the adjusted curve diverging from the findings in the original paper.

Panel B depicts the changes in real log hourly wages when wage growth for service jobs is held constant. In the higher skill percentiles, both the observed and adjusted curves are almost identical, which is reasonable since service jobs are primarily concentrated in the lower skill brackets. In the lower skill percentiles, the adjusted curve aligns more closely with the findings in the original paper. The strong increase in wages for low-skilled jobs is partially explained by the contribution of service jobs, with approximately 50% of the observed wage growth in this segment attributed to service occupations.

Although some differences were identified, the core economic interpretations remain consistent: technological advancements have the strongest impact on non-routine jobs, driving significant changes in both employment share and wages. Non-routine jobs in the middle skill percentiles are displaced by these advancements, while employment and wages for low-skilled manual jobs, such as service occupations, see notable increases.

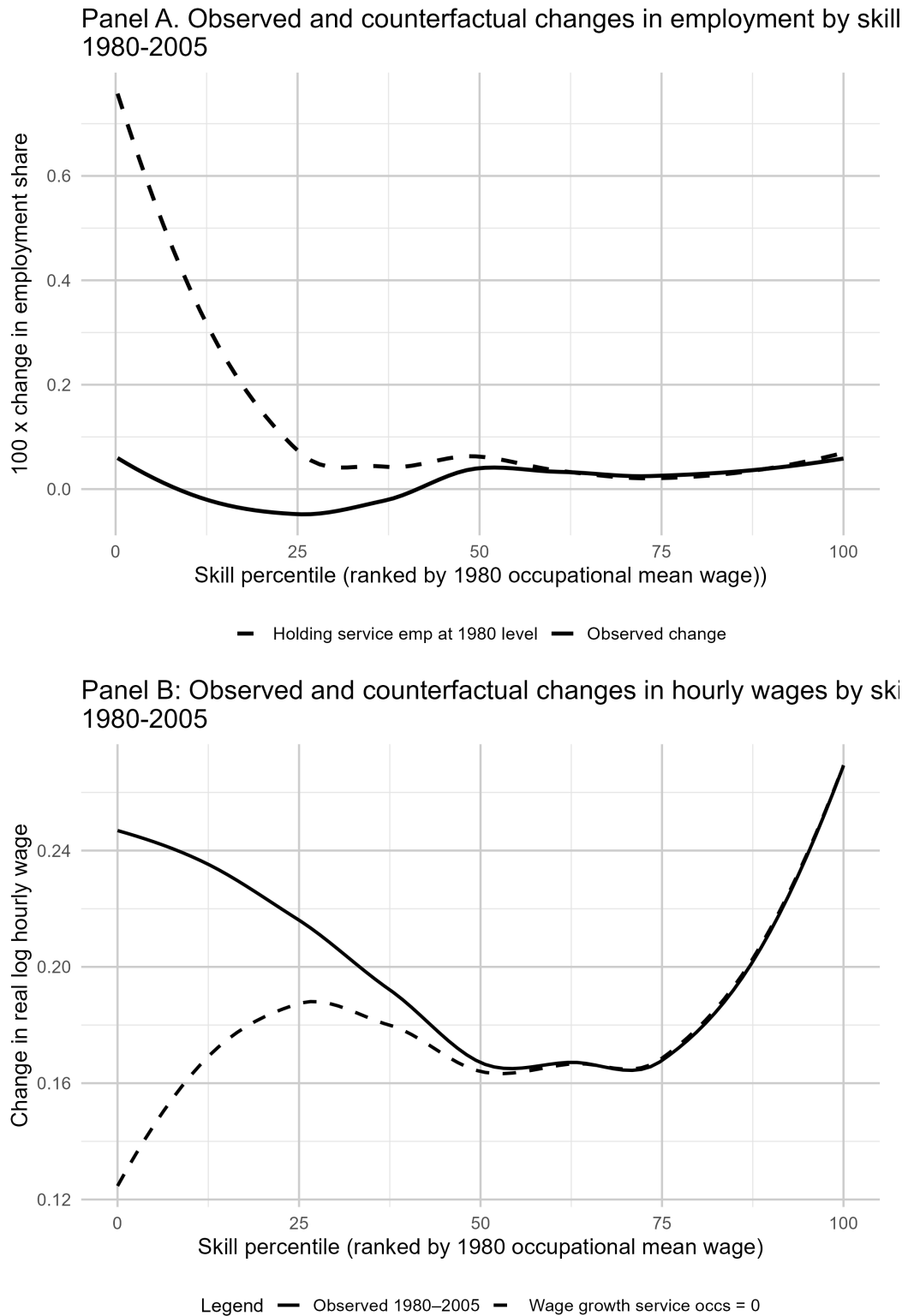


Figure 4: Observed and Counterfactual Changes in Employment and Hourly Wages, 1980–2005